

Aggregate Functions, Conditional Aggregate Functions & Logical Functions

1. Aggregate Functions

What are Aggregate Functions?

Aggregate Functions perform calculations on a group (range) of values and return a **single result**.

They are commonly used to calculate:

- Total Sales
- Average Sales
- Highest Price
- Lowest Price
- Number of Orders

1. SUM Function

Definition

The **SUM** function adds all numeric values within a specified range.

Syntax

=SUM(number1,[number2],...)

or

=SUM(range)

Syntax Explanation

Argument	Meaning
number1	First number or cell reference
number2	Additional numbers or ranges (Optional)
range	Continuous group of cells to be added

Example

Calculate Total Quantity Sold

=SUM(E2:E1002)



Calculate Total Revenue (after Total Amount column is created)

=SUM(H2:H1002)

Returns

The total of all numbers in the selected range.

2. COUNT Function

Definition

The **COUNT** function counts only the cells containing **numeric values** within a specified range.

Syntax

=COUNT(value1,[value2],...)

or

=COUNT(range)

Syntax Explanation

Argument	Meaning
value1	First number or range
value2	Additional values or ranges (Optional)
range	Range containing numeric values

Example

Count Total Orders (based on Quantity)

=COUNT(E2:E1002)

Returns

The total number of cells containing numbers.

Note: COUNT ignores text, blank cells, logical values (TRUE/FALSE), and error values.

COUNTA Function

Definition

The **COUNTA** function counts all **non-empty cells** in a specified range, regardless of the data type.



It counts:

- Numbers
- Text
- Dates
- Time
- Logical Values (TRUE/FALSE)
- Errors

It does **not** count blank cells.

Syntax

=COUNTA(value1,[value2],...)

or

=COUNTA(range)

Syntax Explanation

Argument	Meaning
value1	First value or range
value2	Additional values (Optional)
range	Range to count non-empty cells

Example

Count all Customer Names

=COUNTA(B2:B1002)

Count all Product Entries

=COUNTA(D2:D1002)

Returns

Total number of non-empty cells.

COUNTBLANK Function

Definition



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The **COUNTBLANK** function counts the number of blank (empty) cells within a specified range.

It is commonly used during **data cleaning** to identify missing values.

Syntax

=COUNTBLANK(range)

Syntax Explanation

Argument	Meaning
range	Range where blank cells are counted

Example

Find Missing Customer Names

=COUNTBLANK(B2:B1002)

Find Missing Cities

=COUNTBLANK(C2:C1002)

Returns

Total number of blank cells.

Difference Between COUNT, COUNTA and COUNTBLANK

Function	Counts Numbers	Counts Text	Counts Blank Cells	Counts Dates	Counts TRUE/FALSE
COUNT	☑ Yes	✗ No	✗ No	☑ Yes	✗ No
COUNTA	☑ Yes	☑ Yes	✗ No	☑ Yes	☑ Yes
COUNTBLANK	✗ No	✗ No	☑ Yes	✗ No	✗ No

Business Use Cases

(i) COUNT

Used when counting numeric records.

Examples:

- Number of Orders
- Number of Sales

- Number of Products Sold

Example:

=COUNT(E2:E1002)

(ii) COUNTA

Used when counting completed records.

Examples:

- Total Customer Names
- Total Employee IDs
- Total Products Listed
- Total Records Entered

Example:

=COUNTA(B2:B1002)

(iii) COUNTBLANK

Used during data validation and cleaning.

Examples:

- Missing Customer Names
- Missing Cities
- Missing Product Names
- Incomplete Order Details

Example:

=COUNTBLANK(C2:C1002)

Quick Summary

Function	Purpose
COUNT	Counts only numeric cells.
COUNTA	Counts all non-empty cells, regardless of data type.
COUNTBLANK	Counts only empty (blank) cells.



Interview Tip: A common Excel interview question is "What is the difference between COUNT, COUNTA, and COUNTBLANK?" Remember:

- **COUNT** → Numbers only.
- **COUNTA** → Everything except blanks.
- **COUNTBLANK** → Blank cells only.

3. AVERAGE Function

Definition

Returns the arithmetic mean of selected numbers.

Syntax

=AVERAGE(number1,[number2],...)

or

=AVERAGE(range)

Syntax Explanation

Argument	Meaning
number1	First value
number2	Additional values
range	Range to calculate average

Example

Average Product Price

=AVERAGE(F2:F1002)

Average Quantity Sold

=AVERAGE(E2:E1002)

Returns

Average of all numeric values.

4. MIN Function

Definition

Returns the smallest numeric value from a range.

Syntax

=MIN(number1,[number2],...)

or

=MIN(range)

Example

Lowest Product Price

=MIN(F2:F1002)

Returns

Smallest number in the range.

5. MAX Function

Definition

Returns the largest numeric value from a range.

Syntax

=MAX(number1,[number2],...)

or

=MAX(range)

Example

Highest Product Price

=MAX(F2:F1002)

Returns

Largest number in the range.

Summary of Aggregate Functions

Function	Purpose	Example
SUM	Adds numbers	=SUM(E2:E1002)
COUNT	Counts numeric cells	=COUNT(E2:E1002)
AVERAGE	Calculates average	=AVERAGE(F2:F1002)
MIN	Smallest value	=MIN(F2:F1002)
MAX	Largest value	=MAX(F2:F1002)



2. Conditional Aggregate Functions

What are Conditional Aggregate Functions?

Conditional Aggregate Functions perform calculations only when one or more conditions are satisfied.

Instead of calculating all records, they calculate only matching records.

1. SUMIF Function

Definition

Adds values only when a specified condition is met.

Syntax

`=SUMIF(range,criteria,[sum_range])`

Syntax Explanation

Argument	Meaning
range	Cells where Excel checks the condition
criteria	Condition to satisfy
sum_range	Cells to be added if the condition is TRUE

Example

Total Quantity Sold in Delhi

`=SUMIF(C2:C1002,"Delhi",E2:E1002)`

Total Sales by Rahul Singh

`=SUMIF(G2:G1002,"Rahul Singh",H2:H1002)`

Returns

Sum of matching records.

2. COUNTIF Function

Definition

Counts cells that satisfy a specified condition.



Syntax

=COUNTIF(range,criteria)

Syntax Explanation

Argument	Meaning
range	Cells to evaluate
criteria	Condition to count

Example

Number of Orders from Mumbai

=COUNTIF(C2:C1002,"Mumbai")

Number of Laptop Orders

=COUNTIF(D2:D1002,"Laptop")

Returns

Count of matching records.

3. AVERAGEIF Function

Definition

Calculates the average of values that satisfy a condition.

Syntax

=AVERAGEIF(range,criteria,[average_range])

Syntax Explanation

Argument	Meaning
range	Cells where condition is checked
criteria	Condition
average_range	Values to average

Example

Average Price of Laptops

=AVERAGEIF(D2:D1002,"Laptop",F2:F1002)

Average Quantity Sold in Pune

=AVERAGEIF(C2:C1002,"Pune",E2:E1002)



Returns

Average of matching values.

4. MINIFS Function

Definition

Returns the minimum value that satisfies one or more conditions.

Syntax

=MINIFS(min_range,criteria_range1,criteria1,[criteria_range2,criteria2]...)

Syntax Explanation

Argument	Meaning
min_range	Range from which minimum value is returned
criteria_range1	First range to evaluate
criteria1	First condition
criteria_range2	Additional range (Optional)
criteria2	Additional condition (Optional)

Example

Lowest Laptop Price

=MINIFS(F2:F1002,D2:D1002,"Laptop")

Lowest Quantity Sold in Delhi

=MINIFS(E2:E1002,C2:C1002,"Delhi")

Returns

Smallest matching value.

5. MAXIFS Function

Definition

Returns the largest value that satisfies one or more conditions.

Syntax

=MAXIFS(max_range,criteria_range1,criteria1,[criteria_range2,criteria2]...)

Syntax Explanation

Argument	Meaning
max_range	Range from which maximum value is returned
criteria_range1	First range to evaluate
criterion1	First condition

Example

Highest Laptop Price

=MAXIFS(F2:F1002,D2:D1002,"Laptop")

Highest Quantity in Mumbai

=MAXIFS(E2:E1002,C2:C1002,"Mumbai")

Returns

Largest matching value.

Summary of Conditional Aggregate Functions

Function	Purpose
SUMIF	Sum matching values
COUNTIF	Count matching values
AVERAGEIF	Average matching values
MINIFS	Minimum matching value
MAXIFS	Maximum matching value

3. Logical Functions

What are Logical Functions?

Logical Functions evaluate conditions and return different results based on whether the condition is **TRUE** or **FALSE**.

These functions are widely used for:

- Decision Making
- Data Classification



- Employee Performance
- Student Grades
- Sales Analysis
- Eligibility Checking

1. IF Function

Definition

Checks whether a condition is TRUE or FALSE and returns different results accordingly.

Syntax

=IF(logical_test,value_if_true,value_if_false)

Syntax Explanation

Argument	Meaning
logical_test	Condition to evaluate
value_if_true	Result returned if condition is TRUE
value_if_false	Result returned if condition is FALSE

Example

Suppose orders with Quantity **5 or more** are classified as **Bulk Order**.

=IF(E2>=5,"Bulk Order","Normal Order")

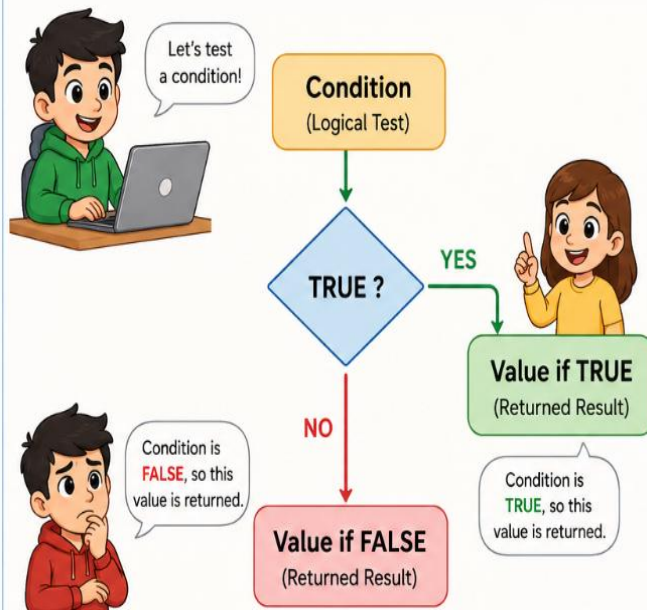
Returns

If Quantity $\geq 5 \rightarrow$ **Bulk Order**

Otherwise $\rightarrow$ **Normal Order**



IF FUNCTION LOGIC FLOW



Syntax:
`=IF(logical_test, value_if_true, value_if_false)`

logical_test → The condition you want to check
value_if_true → The result if the condition is **TRUE**
value_if_false → The result if the condition is **FALSE**

EXAMPLE

Example Scenario

If Quantity is greater than or equal to 5, return "Bulk Order", otherwise return "Normal Order".

Order ID	Customer Name	Quantity	Formula	Result
ORD-100001	Nisha Gupta	5	=IF(E2>=5,"Bulk Order","Normal Order")	Bulk Order
ORD-100002	Deepak Gupta	3	=IF(E3>=5,"Bulk Order","Normal Order")	Normal Order
ORD-100003	Vivaan Sharma	9	=IF(E4>=5,"Bulk Order","Normal Order")	Bulk Order
ORD-100004	Rohit Jain	2	=IF(E5>=5,"Bulk Order","Normal Order")	Normal Order



When the condition is **TRUE**
 (Quantity >= 5)
 Excel returns: **"Bulk Order"**



When the condition is **FALSE**
 (Quantity < 5)
 Excel returns: **"Normal Order"**



This is how **IF function** makes decisions in Excel!

2. Nested IF Function

Definition

A Nested IF is an IF function placed inside another IF function to test multiple conditions.

Syntax

`=IF(logical_test1,value_if_true1,
 IF(logical_test2,value_if_true2,
 value_if_false))`

Example

Classify orders based on Quantity.

`=IF(E2>=8,"High",
 IF(E2>=5,"Medium",
 "Low"))`



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Logic

Quantity	Result
8 or more	High
5-7	Medium
Below 5	Low

NESTED IF LOGIC FLOW



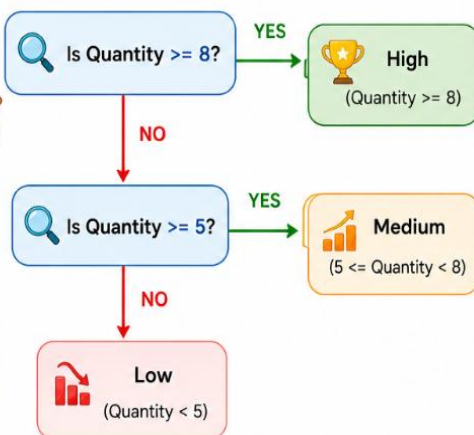
Example Scenario:

Classify orders based on Quantity

High (≥ 8), **Medium** (≥ 5 and < 8), **Low** (< 5)



Think Smart!
Nested IF allows us to test multiple conditions step by step.



Formula (Nested IF):

`=IF(E2>=8,"High",IF(E2>=5,"Medium","Low"))`

First IF checks Quantity ≥ 8

If **FALSE**, it moves to the next IF (Quantity ≥ 5)

If both are **FALSE**, it returns "Low"



EXAMPLE IN DATASET

Order ID	Customer Name	Quantity	Formula Used	Result
ORD-100001	Nisha Gupta	9	<code>=IF(E2>=8,"High",IF(E2>=5,"Medium","Low"))</code>	High
ORD-100002	Deepak Gupta	6	<code>=IF(E3>=8,"High",IF(E3>=5,"Medium","Low"))</code>	Medium
ORD-100003	Vivaan Sharma	3	<code>=IF(E4>=8,"High",IF(E4>=5,"Medium","Low"))</code>	Low
ORD-100004	Rohit Jain	8	<code>=IF(E5>=8,"High",IF(E5>=5,"Medium","Low"))</code>	High



How it Works

- If Quantity $\geq 8 \rightarrow$ **High**
- If Quantity ≥ 5 and $< 8 \rightarrow$ **Medium**
- If Quantity $< 5 \rightarrow$ **Low**



Pro Tip

Use Nested IF when you have multiple conditions to check in a sequence.



Master Logic. Master Excel. Master You!

Practice more, Analyze better, Grow faster!

Difference Between IF and Nested IF

IF	Nested IF
Tests one condition	Tests multiple conditions
Simpler	More complex
Two possible outputs	Multiple possible outputs



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Key Takeaways

- **Aggregate Functions** summarize data by performing calculations on a range of values (e.g., total, average, minimum, maximum).
 - **Conditional Aggregate Functions** perform those calculations only for records that meet specified conditions.
 - **Logical Functions** help make decisions by evaluating conditions and returning different outputs based on the result.
 - Together, these functions form the foundation of business reporting and are among the most frequently used Excel functions in Data Analytics interviews and day-to-day analysis.
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